# Deep Learning for Medical Image Segmentation: A Focus on U-Net for Brain MRI Tumor Detection and Classification


## Developer: Om Patel

## Rocky Point High School


## Introduction

- Medical image segmentation is a crucial task in medical image analysis, which involves identifying and segmenting regions of interest in medical images.
- U-Net is one such approach that has been widely used for medical image segmentation.
- U-Net is a convolutional neural network that was originally introduced for biomedical image processing, where the accuracy of segmentation is very important (Malhotra et al., 2022).
- Image segmentation helps optimize resource allocation by precisely identifying areas that require specific treatments or interventions, leading to more efficient and sustainable practices.
- Deep learning methods have yielded outstanding results in various application domains, including medical image analysis.
- The design of the network structure, 3D data segmentation model design, and loss function design are also important aspects to consider in deep learning approaches for medical image segmentation.
- This research aims to use Computer Vision and Deep Learning to examine splices of Magnetic resonance images and locate tumors autonomously with greater efficiency.
- Image segmentation AI plays a crucial role in the accurate identification and analysis of medical images, aiding in the diagnosis of various conditions such as tumors, lesions, and abnormalities.

## Methods

***1. Software and Hardware setup:***

- To start developing such a program, you will need a few items and software programs. First, you will need a computing device like a laptop or desktop to code on.
- The second thing you will need is Visual Studio Code (VSCode), an IDE allowing users to code which can be installed using this website: https://code.visualstudio.com.
- You will need to install the latest version of Python using this link: https://python.org/.
- Next, you will need a command-line interface, Terminal for Mac, Command Prompt for Windows, or any 3rd party command-line interface like Cmder or Conda will also work. Another item you will need is a web browser that you have unrestricted access to.

***2. Installation and Setup:***

Creating a Visual Studio Code Jupyter project involves several steps to ensure seamless integration of Jupyter notebooks within the VSCode environment:

- Begin by installing Visual Studio Code on your machine using the link above, if you haven't already. Once installed, open VSCode and navigate to the "Extensions" by clicking on the Extensions icon in the Activity Bar on the side of the window.
- Search for the "Jupyter" and "Python" extensions and install it. The Jupyter extension provides Jupyter Notebook support in VSCode and the Python extensions installs all dependencies needed to run Python. After installation, reload the window to activate the extensions. Next, create a new folder on your computer for your Jupyter project, name it something easy to remember and meaningful, and open it in VS Code.
- Then, click on "New File" and if it's an option click on "Jupyter Notebook .ipynb Support". If it's not an option, click on "Python File" and press "Command (Mac) / Control(Windows) + Shift + P" to open up the command palette. In the command palette, type "Create: New Jupyter Notebook" and click it. Now Jupyter Notebook is installed on VSCode.

The next phase in the setup process is to download the brain MRI image slices by navigating to this link: https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation, signing up for a Kaggle account, and downloading all of the files.

*Figure 1* shows the following packages that need to be installed using VSCode in the cells. This should be done by using the "pip install" command. Use the command "!pip freeze" to check the packages that are currently installed. After installing a package, restart the kernel by clicking "restart" next to "Run All".

```
import os
import random
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
plt.style.use("ggplot")
%matplotlib inline

import cv2
from tqdm import tqdm_notebook, trange
from glob2 import glob
from itertools import chain
from skimage.io import imread, imshow, concatenate_images
from skimage.transform import resize
from skimage.morphology import label
from sklearn.model_selection import train_test_split

import tensorflow as tf
from skimage.color import rgb2gray
from tensorflow.keras import Input
from tensorflow.keras.models import Model, load_model, save_model
from tensorflow.keras.layers import Input, Activation, BatchNormalization, Dropout, Lambda, Conv2D, Conv2DTranspose, MaxPooling2D, concatenate
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint

from tensorflow.keras import backend as K
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
```

*Figure 1: shows the following dependencies that will be used for most of the project (two more packages will be installed later because it is not needed at the moment).*

*Figure 2A* shows how to install the packages and important notes. Only some of the packages are shown in *Figure 2A*, so make sure you use the "!pip freeze" command and use Google to install the packages you still need by typing "How to install "package_name" using pip" and clicking on official links like python.org, Stack Overflow, GeeksforGeeks, or other reliable sources. Be sure you see "successfully installed" on the output screen when installing the package, and make sure to restart the kernel after installing each package (red and yellow ovals in *Figure 2A*).

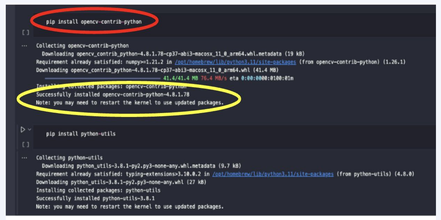



*Figure 2A:* shows the package *OpenCV* and *Utils* being installed in the cells of VSCode.

***3. Finding the dataset***

After installing all the packages, we need to write code to find the Brain MRI slices in your computer (*Figure 3*). The path that is highlighted in yellow in *Figure 3* should be where the MRI slice dataset we downloaded from Kaggle. Find where you installed the dataset, right-click on it, and press "Copy". Then type into the VSCode cell "mask_files = glob(" and then press "Command/ Control + V"). This will paste the path of where the folder is stored on your computer (this path is shown in macOS format and it may look different on windows, but the steps still work). The output should be all the files with the keyword "mask" at the end with the number 3929 which is the correct number of image slices (*Figure 3*).

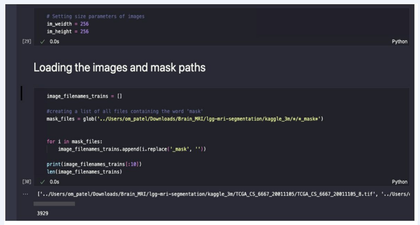



*Figure 3:* shows the code needed to find the slices in your computer

***4. Dice Loss/Coefficient Explanation:***

- The Dice coefficient is a measure of the similarity between two samples.
- In the context of image segmentation, it is used to evaluate the performance of an algorithm by comparing the overlap between the predicted and ground truth segmentation masks.
- The Dice coefficient ranges from 0 to 1, where 1 indicates perfect overlap and 0 indicates no overlap.
- In the context of image segmentation, the predicted mask and ground truth mask (Table 1) refer to binary images that represent the regions of interest in an image (*Figure 4A*).
- The model is presented with input images, and the loss between the predicted masks and the ground truth masks is computed using metrics like the Dice coefficient or other segmentation loss functions.
- The Dice loss is a loss function derived from the Dice coefficient.
- Minimizing the Dice loss during training encourages the model to produce segmentation masks that closely match the ground truth (*Figure 4B*).
- The closer the Dice Loss is to 0, the better the alignment between the predicted and ground truth masks.

| *Ground Truth Mask:* | *Predicted Mask:* |
|---|---|
| The ground truth mask is the manually annotated or labeled segmentation mask that indicates the correct segmentation of objects or regions in an image. | -The predicted mask is the output of a segmentation model, representing the model's estimation of the regions or objects in the input image. |
| Each pixel in the ground truth mask is assigned a label, typically 1 for the object of interest and 0 for the background. | Similar to the ground truth mask, the predicted mask is a binary image where each pixel is assigned a label, with 1 indicating the predicted object and 0 indicating the predicted background. |
| Ground truth masks are used during the training of a segmentation model to provide a target for the model to learn from. | The goal during training is to optimize the model so that its predicted masks closely match the ground truth masks. |

*(Table 1):* shows the main differences between the term *ground truth mask* and *predicted mask*.

$$Dice(X,Y) = \frac{2 * |X \cap Y|}{|X| + |Y|}$$

*(Figure 4A):* shows the formula for the Dice coefficient where X is the predicted binary mask. Y is the ground truth binary mask. $|X \cap Y|$ is the intersection of the predicted and ground truth masks. $|X|$ is the total number of pixels in the predicted mask. $|Y|$ is the total number of pixels in the ground truth mask.

$$DiceLoss(X,Y) = 1 - \frac{2 * |X \cap Y|}{|X| + |Y|} \quad \text{or} \quad Dice_Loss(X,Y) = 1 - Dice(X,Y)$$

*(Figure 4B):* shows the dice loss function, and the variable is the same as the Dice Coefficient, but with a the Dice Coefficient being subtracted from 1

***5. U-Net Explanation:***

U-Net Explanation:

- U-Net is a convolutional neural network architecture used for image segmentation.
- It has a distinct U-shaped design with key components.

Encoder (Contracting Path):

- Located at the top of the U-shaped design.
- Comprises convolutional and pooling layers.
- Uses small receptive field convolutions to capture features in the input image.

Decoder (Expanding Path):

- Constitutes the bottom part of the architecture.
- Includes up-sampling and concatenation operations.
- Up-sampling increases the spatial resolution of feature maps.
- Concatenation integrates low-level and high-level features.

Skip Connections:

- Key innovation linking corresponding feature maps from the encoder to the decoder.
- Acts as a shortcut for gradient flow during backpropagation.
- Preserves fine-grained details from early layers, mitigating the vanishing gradient problem.

Final Layer:

- Typically consists of a convolutional layer with a sigmoid activation function.
- Network's output is a pixel-wise segmentation mask.
- Each pixel is assigned a probability value indicating the likelihood of belonging to the object of interest.

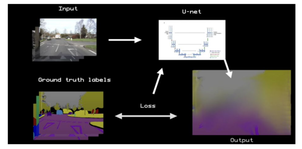
Training Process:

- Involves feeding input images into the network.
- Obtaining predicted masks and comparing them with ground truth masks.
- Uses Dice Loss to minimize during training.
- Goal is to produce accurate and precise segmentation masks.

Python Package:

- To use U-Net, download the Python package "unet."

The goal is to minimize the Dice Loss during training, which encourages the U-Net to produce accurate and precise segmentation masks that closely match the ground truth. Figure 5 explains how U-Net is incorporated into the program *(Figure 5)*.




*(Figure 5):* shows the training process of U-Net goes through from each image.

**Note that this image was from a youtube video and cited in the *References*

## Results

The output *(Figure 6)* shows the training process of a convolutional neural network (CNN) using the U-Net architecture for image segmentation. The training is performed for 100 epochs, where each epoch involves passing the entire dataset through the network once.

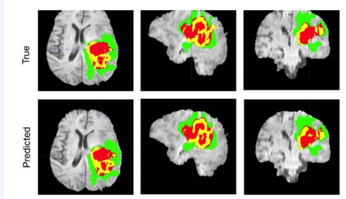
*(Figure 6): Shows the output of when running the cell of the Training Parameters of the AI segmentation*

1. A total of 2,828 validated image filenames were found, which indicates the number of images used for training and validation.
2. The model's loss, binary accuracy, Intersection over Union (IoU), and the Dice coefficient are reported for both training and validation sets. Loss is a measure of the difference between predicted and actual values, while binary accuracy calculates the proportion of correct predictions. IoU and the Dice coefficient are measures of overlap between the predicted and ground truth segmentation masks.
3. The model reports improved validation loss (-0.02516) after the first epoch, indicating better performance on the validation set compared to the initial loss. The improved validation loss triggers the model to save itself to a file named "unet.hdf5".
4. The training and validation loss, binary accuracy, IoU, and the Dice coefficient are reported for each epoch. As the training progresses, the model's performance on the training set improves, as indicated by the decreasing loss and increasing binary accuracy, IoU, and the Dice coefficient.
5. After 10 epochs, the training loss is -0.1960, binary accuracy is 0.9833, IoU is 0.1103, and the Dice coefficient is 0.1960. These results suggest that the model has achieved reasonable performance on the training set.
6. A UserWarning is displayed, suggesting that the HDF5 file format used for saving the model is considered legacy. The warning recommends using the native Keras format instead

## Discussion

While creating the program, the biggest issue that came across was when running the training parameters and training the AI. It took 40 minutes to complete one epoch which would mean that it would take 5 ½ days to run at the ideal epoch of 200. This problem could be resolved by upgrading the hardware on my computer. Image segmentation and Computer Vision are not fully supported and are rather slow on MacOS. This is due to the M1/M2 chip processor not fully supported for Tensorflow and with the optimizers that were used in this program. Mostafa's article mentioned in the paper managed to run the program within a few seconds. My rationale behind why he was able to train the AI quickly was because he either has a powerful Windows PC with expensive specs or he was using the newest optimizers and package versions that may work for his situation but did not work for me because Python packages and optimizers change weekly or monthly and depreciate, resulting into different results. The expected outcome was that it would take the true images from Kaggle, compare them to the user's images, and predict where the tumor is located based on the true image (*Figure 7*). The use of computer vision and AI in medical image segmentation, particularly in the detection of tumors, holds significant implications for healthcare. Deep learning approaches, such as U-Net, have shown promising results in accurately identifying and segmenting regions of tumors in brain MRI images. The integration of computer vision and deep learning aims to create autonomous programs for tumor detection in MRI images, offering potential benefits in early diagnosis and personalized treatment plans. Overall, these advancements in medical image segmentation contribute to the broader goal of improving accuracy, efficiency, and personalized healthcare through the synergy of computer vision and AI in medical imaging.

Image taken from: *https://developer-blogs.nvidia.com/wp-content/uploads/2018/11/Screen-Shot-2018-11-26-at-8.36.35-PM.png*




*(Figure 7): This is the result that would have shown if the training ran at a reasonable rate. It shows the true image that the AI uses, and the tumor predicted image, where the tumor that is predicted is in yellow.*

## Conclusion

The objective of this research is to develop an AI program capable of segmenting and detecting tumors in Brain MRI scans, with an additional focus on enhancing the speed of the process. There have been suggestions for annotation-efficient deep-learning frameworks to enhance the diagnosis of medical images. The automation and acceleration of the segmentation process in medical imaging can be achieved through the potential of deep learning approaches. The identification and segmentation of crucial areas in medical images, such as tumors and lesions, are imperative in medical image segmentation. Obstacles encompass the requirement for an ample amount of training data, the enhancement of preprocessing techniques, and the resolution of class imbalance. The effectiveness of deep learning models is significantly influenced by the caliber of the training datasets. The main limitations encountered in the development of the program include the time-intensive training process, taking 40 minutes for one epoch, leading to a projected 5 ½ days for 200 epochs. Compatibility issues on MacOS, specifically with M1/M2 processors, contribute to slow image segmentation and Computer Vision due to incomplete TensorFlow support. The reliance on hardware and software versions, along with the dynamic nature of Python packages and optimizers, poses challenges in achieving consistent results. The author expresses a future aspiration to use a more powerful PC for quicker program execution and efficient exploration of medical image segmentation and AI applications. The overall significance of my study is to better understand the area of Image Segmentation and what is supported and what is not supported with a given piece of hardware. It also gave further insight of what optimizers and other packages future developers can use. In the future, a significantly more powerful PC than a macbook should be used to run the program in less time. The reason for this is so the images of the segmented areas of the tumors can be made.

## References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All *Figures* were created and taken by Om Patel except for *Figures 5 and 7 (links to origin of picture are provided)*


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